Ubuntu Server Deployment Report

Installation • Network Configuration • Static IP • OpenSSH • Remote Access

Virtual machine	Ubuntu-Server-M009	**Operating system**	Ubuntu Server 26.04.1 LTS
Administrator account	huzaifa	**Report status**	Core deployment and SSH validation completed

1. Executive Summary

This report documents the deployment and initial configuration of an Ubuntu Server virtual machine for a cybersecurity learning environment. The VM was installed in Oracle VirtualBox, configured with NAT, internal and host-only networking, assigned a persistent host-only IPv4 address using Netplan, and enabled for remote administration through OpenSSH. A successful SSH login from Windows PowerShell verified connectivity and authentication.

> **Outcome:** The server is reachable from the Windows host at **192.168.220.30** over the VirtualBox host-only network, while its NAT interface retains the address **10.0.2.15** for outbound connectivity.

2. Objectives

1. Install Ubuntu Server in a VirtualBox virtual machine.
2. Identify the available network interfaces.
3. Configure a persistent static IP address.
4. Update package information and installed packages.
5. Install and enable OpenSSH Server.
6. Test remote login from Windows.
7. Prepare a deployment record.

3. Environment and Network Design

The deployment used Oracle VirtualBox on a Windows host. The virtual machine specification and network adapter layout are summarised below and shown in Figure 1.

Virtual machine configuration

Component	Configuration
Processors	2
Boot and acceleration	Boot order: hard disk, optical, floppy; nested paging and KVM paravirtualization
Display	VMSVGA graphics controller, 16 MB video memory; remote desktop server and recording disabled
Storage	SATA port 0: Ubuntu-Server-M009.vdi (normal, 25.03 GB); IDE optical drive empty
Audio and USB	Default host driver, ICH AC97 controller; OHCI and EHCI USB controllers, no device filters

Network adapters

Interface	VirtualBox adapter	Observed address	Purpose
enp0s3	Adapter 1: NAT	10.0.2.15/24	Outbound connectivity; default route via 10.0.2.2
enp0s8	Adapter 2: Internal Network (SOC-INTERNAL)	No IPv4 address observed	Isolated lab segment; not used for the static host-only connection
enp0s9	Adapter 3: Host-only (VirtualBox Host-Only Ethernet Adapter #2)	192.168.220.30/24	Private link between the Windows host and the Ubuntu VM
Windows host adapter	Host-only	192.168.220.1/24	Windows endpoint for host-only communication

All three adapters use the Intel PRO/1000 MT Desktop model.

Network rationale. NAT gives the VM a path to external networks through VirtualBox. The internal network (SOC-INTERNAL) provides an isolated segment for future lab machines. The host-only adapter creates a private network shared with the Windows host, allowing SSH access without exposing the server on the physical LAN. The host-only interface has no default gateway configured; the default route remains through NAT.

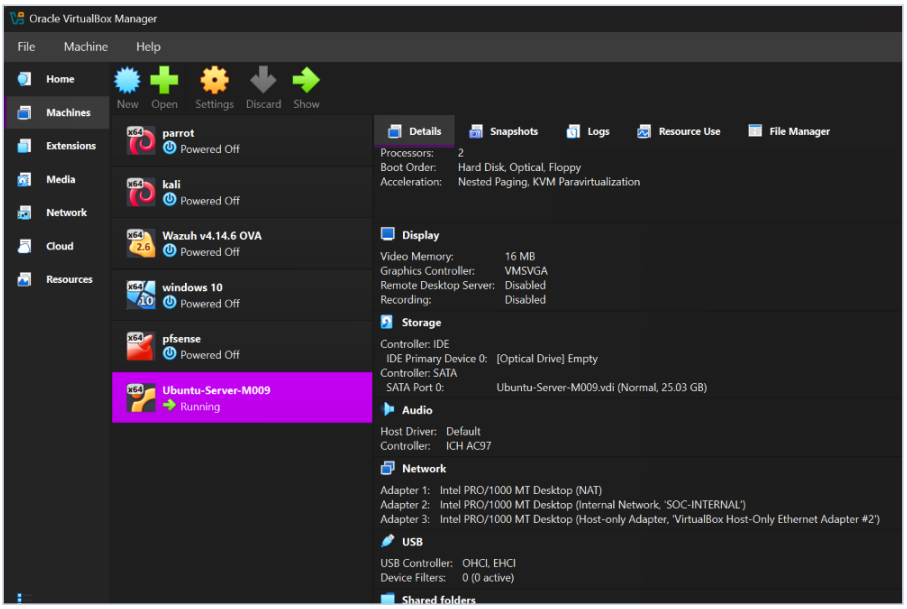


Figure 1. VirtualBox Manager showing the Ubuntu-Server-M009 hardware and network adapter configuration.

4. Network Discovery and Static IP Configuration

The interfaces were inspected with `ip -br addr` and routing was checked with `ip route`. The existing Netplan file, located at `/etc/netplan/00-installer-config.yaml`, was edited so that `enp0s9` uses a fixed IPv4 address instead of DHCP.

Commands used

```
ip -br addr
ip route
ls -l /etc/netplan
sudo cat /etc/netplan/00-installer-config.yaml
sudo netplan get
sudo nano /etc/netplan/00-installer-config.yaml
sudo netplan generate
sudo netplan try
```

Final Netplan configuration

```
network:
  version: 2
  ethernets:
    enp0s3:
      dhcp4: true
      dhcp6: true
      match:
        macaddress: "08:00:27:0e:59:2d"
      set-name: enp0s3
    enp0s8:
      accept-ra: true
    enp0s9:
      dhcp4: false
      dhcp6: false
      accept-ra: false
      addresses:
        - 192.168.220.30/24
```

Validation. The configuration was checked with `sudo netplan generate`, which returned no errors, and applied with `sudo netplan try`. A follow-up address check confirmed that `enp0s9` held 192.168.220.30/24.

5. Package Updates and OpenSSH Installation

Package metadata and available updates were refreshed, and OpenSSH Server was installed to provide encrypted remote administration.

Commands used

```
sudo apt update
sudo apt upgrade
sudo apt install openssh-server
sudo systemctl enable --now ssh
sudo systemctl status ssh
systemctl is-active ssh
```

> **Validation result:** `active`. The SSH service is enabled to start automatically at boot and was confirmed to be running (Figure 2).

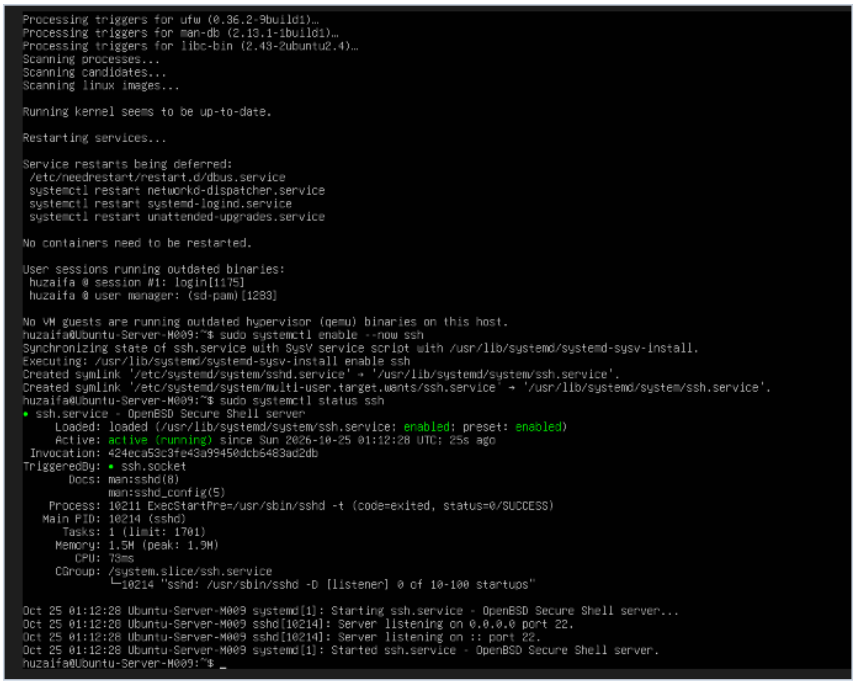


Figure 2. OpenSSH service enabled and active (running), listening on port 22.

6. Remote Access Test from Windows

The following command was executed from Windows PowerShell:

```
ssh huzaifa@192.168.220.30
```

On first connection, SSH displayed the ED25519 host-key fingerprint prompt. The key was accepted, the account password was entered, and the Ubuntu Server login banner and shell prompt were returned. The session verified host reachability, SSH service availability, and user authentication. The verification output is shown in Figure 3.

Remote verification commands

```
hostname
whoami
ip -br addr
systemctl is-active ssh
exit
```

Observed results

Check	Observed result
Hostname	Ubuntu-Server-M009
Logged-in user	huzaifa
Host-only IPv4	enp0s9 - 192.168.220.30/24
NAT IPv4	enp0s3 - 10.0.2.15/24
SSH service state	active
SSH session	Successful; clean logout returned to PowerShell



Figure 3. Verification commands run over the SSH session from Windows PowerShell, ending with a clean logout.

7. Security and Operational Notes

- Keep the host-only network for private management access wherever possible.
- Use strong, unique passwords, and consider SSH key-based authentication as a future hardening step.
- Restrict SSH exposure with firewall rules when adding network interfaces or services.
- Apply security updates regularly and review authentication logs when investigating access attempts.
- Before changing Netplan remotely, validate with `sudo netplan generate` and use `sudo netplan try` to reduce the risk of losing connectivity.

8. Snapshot and Deliverables

Recommended snapshot name: MISSION-009-SSH-Configured

Suggested description: Ubuntu Server installed, static host-only IP configured, and SSH login verified from Windows PowerShell.

> **Status note:** The snapshot was recommended in the workflow, but its creation has not been confirmed. Create the snapshot before marking this item complete.

Documentation deliverable: this report (PDF).

9. Conclusion

Ubuntu Server was deployed in VirtualBox and configured for reliable private remote management. Netplan provided a persistent static host-only address, while NAT remained the default route for outbound connectivity. OpenSSH was installed, enabled, and verified as active, and a successful Windows PowerShell session confirmed that the server can be administered remotely. The deployment provides a foundation for future Linux administration, server hardening, web hosting, and controlled cybersecurity lab exercises.

10. References

- Ubuntu Server documentation: https://ubuntu.com/server/docs/
- Ubuntu networking and Netplan: https://ubuntu.com/server/docs/configuring-networks/
- Ubuntu OpenSSH Server guide: https://ubuntu.com/server/docs/how-to/security/openssh-server/

Prepared from the configuration and command outputs recorded during the deployment. Values are specific to this virtual lab and may change if the VM or VirtualBox network settings are modified.